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1. The *Catalan numbers* are given by

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

for any integer $n \geq 0$. (Recall that $\binom{0}{0} = 1$ – there’s only one way to choose no elements from an empty set.) Prove that

$$C_n = \frac{2(2n-1)}{n+1} C_{n-1}$$

for all $n \geq 1$, and hence that $C_n \leq 4^n$ for all $n \geq 0$.

Solution: We write

$$\begin{aligned} C_n &= \frac{1}{n+1} \binom{2n}{n} \\ &= \frac{1}{n+1} \frac{(2n)!}{n!n!} \\ &= \frac{(2n)(2n-1)}{(n+1)(n)(n)} \frac{(2n-2)!}{(n-1)!(n-1)!} \\ &= \frac{2(2n-1)}{n+1} \frac{1}{n} \binom{2(n-1)}{n-1} \\ &= \frac{2(2n-1)}{n+1} C_{n-1}. \end{aligned}$$

This proves the first part. For the second part, we note simply that

$$\frac{2(2n-1)}{n+1} < \frac{2(2n+2)}{n+1} = 4.$$

Since $C_0 = 1$, the rest follows immediately by induction.

2. Let m, n be positive integers. A function $f : \{1, \dots, m\} \rightarrow \{1, \dots, n\}$ is said to be *weakly increasing* if $x < y$ implies $f(x) \leq f(y)$. For fixed m, n , how many weakly increasing functions are there? Write your answer as an expression involving m, n and justify how you derived that expression.

Solution: For a given increasing function, since its outputs are ordered:

$$f(1) \leq f(2) \leq \dots \leq f(m),$$

then f is completely determined by its multi-set of outputs. Hence all we need to know is how often f takes on each particular output $i \in \{1, \dots, n\}$. Defining $x_i := |f^{-1}(i)|$ as the number of times f takes output i , we can see:

$$x_1 + x_2 + \dots + x_n = m.$$

with the constraint $x_i \geq 0$. Hence this becomes a standard stars and bars problem where we divide m objects into n groups. There are $\binom{n+m-1}{n-1}$ of doing this.